5

Stability and Performance of Feedback Systems

This chapter introduces the feedback structure and discusses its stability and performance properties. The arrangement of this chapter is as follows: Section 5.1 discusses the necessity for introducing feedback structure and describes the general feedback configuration. In section 5.2, the well-posedness of the feedback loop is defined. Next, the notion of internal stability is introduced and the relationship is established between the state space characterization of internal stability and the transfer matrix characterization of internal stability in section 5.3. The stable coprime factorizations of rational matrices are also introduced in section 5.4. Section 5.5 considers feedback properties and discusses how to achieve desired performance using feedback control. These discussions lead to a loop shaping control design technique which is introduced in section 5.6. Finally, we consider the mathematical formulations of optimal $\mathcal{H}_2$ and $\mathcal{H}_\infty$ control problems in section 5.7.

5.1 Feedback Structure

In designing control systems, there are several fundamental issues that transcend the boundaries of specific applications. Although they may differ for each application and may have different levels of importance, these issues are generic in their relationship to control design objectives and procedures. Central to these issues is the requirement to provide satisfactory performance in the face of modeling errors, system variations, and uncertainty. Indeed, this requirement was the original motivation for the development of feedback systems. Feedback is only required when system performance cannot be achieved because of uncertainty in system characteristics. The more detailed treatment of model uncertainties and their representations will be discussed in Chapter 9.

For the moment, assuming we are given a model including a representation of uncertainty which we believe adequately captures the essential features of the plant, the next step in the controller design process is to determine what structure is necessary to achieve the desired performance. Prefiltering input signals (or open loop control) can change the dynamic response of the model set but cannot reduce the effect of uncertainty. If the uncertainty is too great to achieve the desired accuracy of response, then a feedback structure is required. The mere assumption of a feedback structure, however, does not guarantee a reduction of uncertainty, and there are many obstacles to achieving the uncertainty-reducing benefits of feedback. In particular, since for any reasonable model set representing a physical system uncertainty becomes large and the phase is completely unknown at sufficiently high frequencies, the loop gain must be small at those frequencies to avoid destabilizing the high frequency system dynamics. Even worse is that the feedback system actually increases uncertainty and sensitivity in the frequency ranges where uncertainty is significantly large. In other words, because of the type of sets required to reasonably model physical systems and because of the restriction that our controllers be causal, we cannot use feedback (or any other control structure) to cause our closed-loop model set to be a proper subset of the open-loop model set. Often, what can be achieved with intelligent use of feedback is a significant reduction of uncertainty for certain signals of importance with a small increase spread over other signals. Thus, the feedback design problem centers around the tradeoff involved in reducing the overall impact of uncertainty. This tradeoff also occurs, for example, when using feedback to reduce command/disturbance error while minimizing response degradation due to measurement noise. To be of practical value, a design technique must provide means for performing these tradeoffs. We will discuss these tradeoffs in more detail later in section 5.5 and in Chapter 6.

To focus our discussion, we will consider the standard feedback configuration shown in Figure 5.1. It consists of the interconnected plant P and controller K forced by

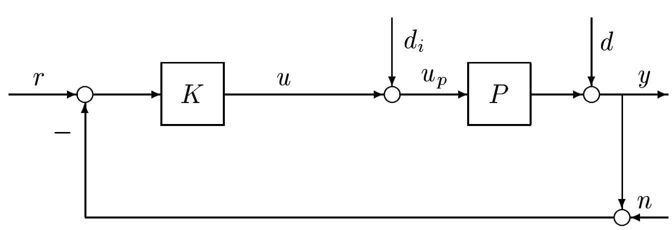

Figure 5.1: Standard Feedback Configuration

command r, sensor noise n, plant input disturbance d_i, and plant output disturbance d. In general, all signals are assumed to be multivariable, and all transfer matrices are assumed to have appropriate dimensions.

5.2 Well-Posedness of Feedback Loop

Assume that the plant P and the controller K in Figure 5.1 are fixed real rational proper transfer matrices. Then the first question one would ask is whether the feedback interconnection makes sense or is physically realizable. To be more specific, consider a simple example where

$$P = -\frac{s-1}{s+2}, \qquad K = 1$$

are both proper transfer functions. However,

$$u = \frac{(s+2)}{3}(r-n-d) - \frac{s-1}{3}d_i$$

i.e., the transfer functions from the external signals $r-n-d$ and d_i to u are not proper. Hence, the feedback system is not physically realizable!

Definition 5.1 A feedback system is said to be *well-posed* if all closed-loop transfer matrices are well-defined and proper.

Now suppose that all the external signals r, n, d, and d_i are specified and that the closed-loop transfer matrices from them to u are respectively well-defined and proper. Then, y and all other signals are also well-defined and the related transfer matrices are proper. Furthermore, since the transfer matrices from d and n to u are the same and differ from the transfer matrix from r to u by only a sign, the system is well-posed if and only if the transfer matrix from $\begin{bmatrix} d_i \\ d \end{bmatrix}$ to u exists and is proper.

In order to be consistent with the notation used in the rest of the book, we shall denote

$$\hat{K} := -K$$

and regroup the external input signals into the feedback loop as w_1 and w_2 and regroup the input signals of the plant and the controller as e_1 and e_2. Then the feedback loop with the plant and the controller can be simply represented as in Figure 5.2 and the system is well-posed if and only if the transfer matrix from $\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ to e_1 exists and is proper.

Lemma 5.1 *The feedback system in Figure 5.2 is well-posed if and only if*

$$I - \hat{K}(\infty)P(\infty) \tag{5.1}$$

is invertible.

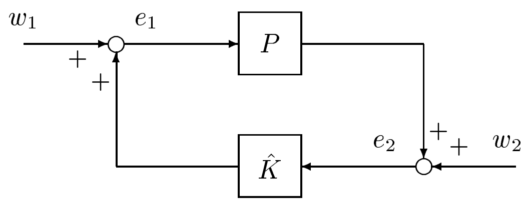

Figure 5.2: Internal Stability Analysis Diagram

Proof. The system in the above diagram can be represented in equation form as

$$\begin{aligned} e_1 &= w_1 + \hat{K}e_2 \\ e_2 &= w_2 + Pe_1. \end{aligned}$$

Then an expression for e_1 can be obtained as

$$(I - \hat{K}P)e_1 = w_1 + \hat{K}w_2.$$

Thus well-posedness is equivalent to the condition that $(I - \hat{K}P)^{-1}$ exists and is proper. But this is equivalent to the condition that the constant term of the transfer function $I - \hat{K}P$ is invertible. $\square$

It is straightforward to show that (5.1) is equivalent to either one of the following two conditions:

$$\begin{bmatrix} I & -\hat{K}(\infty) \\ -P(\infty) & I \end{bmatrix} \text{ is invertible;} \tag{5.2}$$

$$I - P(\infty)\hat{K}(\infty) \text{ is invertible.}$$

The well-posedness condition is simple to state in terms of state-space realizations. Introduce realizations of P and $\hat{K}$:

$$P = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] \tag{5.3}$$

$$\hat{K} = \left[\begin{array}{c|c} \hat{A} & \hat{B} \\ \hline \hat{C} & \hat{D} \end{array}\right]. \tag{5.4}$$

Then $P(\infty) = D$ and $\hat{K}(\infty) = \hat{D}$. For example, well-posedness in (5.2) is equivalent to the condition that

$$\begin{bmatrix} I & -\hat{D} \\ -D & I \end{bmatrix} \text{ is invertible.} \tag{5.5}$$

Fortunately, in most practical cases we will have $D = 0$, and hence well-posedness for most practical control systems is guaranteed.

5.3 Internal Stability

Consider a system described by the standard block diagram in Figure 5.2 and assume the system is well-posed. Furthermore, assume that the realizations for $P(s)$ and $\hat{K}(s)$ given in equations (5.3) and (5.4) are *stabilizable and detectable.*

Let x and $\hat{x}$ denote the state vectors for P and $\hat{K}$, respectively, and write the state equations in Figure 5.2 with w_1 and w_2 set to zero:

$$\begin{aligned} \dot{x} &= Ax + Be_1 && (5.6)\\ e_2 &= Cx + De_1 && (5.7)\\ \dot{\hat{x}} &= \hat{A}\hat{x} + \hat{B}e_2 && (5.8)\\ e_1 &= \hat{C}\hat{x} + \hat{D}e_2. && (5.9) \end{aligned}$$

Definition 5.2 The system of Figure 5.2 is said to be *internally stable* if the origin $(x, \hat{x}) = (0, 0)$ is asymptotically stable, i.e., the states $(x, \hat{x})$ go to zero from all initial states when $w_1 = 0$ and $w_2 = 0$.

Note that internal stability is a state space notion. To get a concrete characterization of internal stability, solve equations (5.7) and (5.9) for e_1 and e_2:

$$\begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} I & -\hat{D} \\ -D & I \end{bmatrix}^{-1} \begin{bmatrix} 0 & \hat{C} \\ C & 0 \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix}.$$

Note that the existence of the inverse is guaranteed by the well-posedness condition. Now substitute this into (5.6) and (5.8) to get

$$\begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} = \tilde{A} \begin{bmatrix} x \\ \hat{x} \end{bmatrix}$$

where

$$\tilde{A} = \begin{bmatrix} A & 0 \\ 0 & \hat{A} \end{bmatrix} + \begin{bmatrix} B & 0 \\ 0 & \hat{B} \end{bmatrix} \begin{bmatrix} I & -\hat{D} \\ -D & I \end{bmatrix}^{-1} \begin{bmatrix} 0 & \hat{C} \\ C & 0 \end{bmatrix}.$$

Thus internal stability is equivalent to the condition that $\tilde{A}$ has all its eigenvalues in the open left-half plane. In fact, this can be taken as a definition of internal stability.

Lemma 5.2 *The system of Figure 5.2 with given stabilizable and detectable realizations for P and $\hat{K}$ is internally stable if and only if $\tilde{A}$ is a Hurwitz matrix.*

It is routine to verify that the above definition of internal stability depends only on P and $\hat{K}$, not on specific realizations of them as long as the realizations of P and $\hat{K}$ are both stabilizable and detectable, i.e., no extra unstable modes are introduced by the realizations.

The above notion of internal stability is defined in terms of state-space realizations of P and $\hat{K}$. It is also important and useful to characterize internal stability from the transfer matrix point of view. Note that the feedback system in Figure 5.2 is described, in term of transfer matrices, by

$$\begin{bmatrix} I & -\hat{K} \\ -P & I \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}. \tag{5.10}$$

Now it is intuitively clear that if the system in Figure 5.2 is internally stable, then for all bounded inputs (w_1, w_2), the outputs (e_1, e_2) are also bounded. The following lemma shows that this idea leads to a transfer matrix characterization of internal stability.

Lemma 5.3 *The system in Figure 5.2 is internally stable if and only if the transfer matrix*

$$\begin{bmatrix} I & -\hat{K} \\ -P & I \end{bmatrix}^{-1} = \begin{bmatrix} I + \hat{K}(I - P\hat{K})^{-1}P & \hat{K}(I - P\hat{K})^{-1} \\ (I - P\hat{K})^{-1}P & (I - P\hat{K})^{-1} \end{bmatrix} \tag{5.11}$$

from (w_1, w_2) *to* (e_1, e_2) *belongs to* $\mathcal{RH}_\infty$.

Proof. As above let $\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ and $\left[\begin{array}{c|c} \hat{A} & \hat{B} \\ \hline \hat{C} & \hat{D} \end{array}\right]$ be stabilizable and detectable realizations of P and $\hat{K}$, respectively. Let y_1 denote the output of P and y_2 the output of $\hat{K}$. Then the state-space equations for the system in Figure 5.2 are

$$\begin{aligned}
\begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} &= \begin{bmatrix} A & 0 \\ 0 & \hat{A} \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} + \begin{bmatrix} B & 0 \\ 0 & \hat{B} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} \\
\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} &= \begin{bmatrix} C & 0 \\ 0 & \hat{C} \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} + \begin{bmatrix} D & 0 \\ 0 & \hat{D} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} \\
\begin{bmatrix} e_1 \\ e_2 \end{bmatrix} &= \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} + \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}.
\end{aligned}$$

The last two equations can be rewritten as

$$\begin{bmatrix} I & -\hat{D} \\ -D & I \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 0 & \hat{C} \\ C & 0 \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}.$$

Now suppose that this system is internally stable. Then the well-posedness condition implies that $(I - D\hat{D}) = (I - P\hat{K})(\infty)$ is invertible. Hence, $(I - P\hat{K})$ is invertible. Furthermore, since the eigenvalues of

$$\tilde{A} = \begin{bmatrix} A & 0 \\ 0 & \hat{A} \end{bmatrix} + \begin{bmatrix} B & 0 \\ 0 & \hat{B} \end{bmatrix} \begin{bmatrix} I & -\hat{D} \\ -D & I \end{bmatrix}^{-1} \begin{bmatrix} 0 & \hat{C} \\ C & 0 \end{bmatrix}$$

are in the open left-half plane, it follows that the transfer matrix from (w_1, w_2) to (e_1, e_2) given in (5.11) is in $\mathcal{RH}_\infty$.

Conversely, suppose that $(I - P\hat{K})$ is invertible and the transfer matrix in (5.11) is in $\mathcal{RH}_\infty$. Then, in particular, $(I - P\hat{K})^{-1}$ is proper which implies that $(I - P\hat{K})(\infty) = (I - D\hat{D})$ is invertible. Therefore,

$$\begin{bmatrix} I & -\hat{D} \\ -D & I \end{bmatrix}$$

is nonsingular. Now routine calculations give the transfer matrix from $\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ to $\begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$ in terms of the state space realizations:

$$\begin{bmatrix} I & -\hat{D} \\ -D & I \end{bmatrix}^{-1} \left\{ \begin{bmatrix} I & -\hat{D} \\ -D & I \end{bmatrix} + \begin{bmatrix} 0 & \hat{C} \\ C & 0 \end{bmatrix} (sI - \tilde{A})^{-1} \begin{bmatrix} B & 0 \\ 0 & \hat{B} \end{bmatrix} \right\} \begin{bmatrix} I & -\hat{D} \\ -D & I \end{bmatrix}^{-1}.$$

Since the above transfer matrix belongs to $\mathcal{RH}_\infty$, it follows that

$$\begin{bmatrix} 0 & \hat{C} \\ C & 0 \end{bmatrix} (sI - \tilde{A})^{-1} \begin{bmatrix} B & 0 \\ 0 & \hat{B} \end{bmatrix}$$

as a transfer matrix belongs to $\mathcal{RH}_\infty$. Finally, since (A, B, C) and $(\hat{A}, \hat{B}, \hat{C})$ are stabilizable and detectable,

$$\left(\tilde{A}, \begin{bmatrix} B & 0 \\ 0 & \hat{B} \end{bmatrix}, \begin{bmatrix} 0 & \hat{C} \\ C & 0 \end{bmatrix} \right)$$

is stabilizable and detectable. It then follows that the eigenvalues of $\tilde{A}$ are in the open left-half plane. $\square$

Note that to check internal stability, it is necessary (and sufficient) to test whether each of the four transfer matrices in (5.11) is in $\mathcal{RH}_\infty$. Stability cannot be concluded even if three of the four transfer matrices in (5.11) are in $\mathcal{RH}_\infty$. For example, let an interconnected system transfer function be given by

$$P = \frac{s-1}{s+1}, \qquad \hat{K} = -\frac{1}{s-1}.$$

Then it is easy to compute

$$\begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} \dfrac{s+1}{s+2} & -\dfrac{s+1}{(s-1)(s+2)} \\ \dfrac{s-1}{s+2} & \dfrac{s+1}{s+2} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix},$$

which shows that the system is not internally stable although three of the four transfer functions are stable. This can also be seen by calculating the closed-loop A-matrix with any stabilizable and detectable realizations of P and $\hat{K}$.

Remark 5.1 It should be noted that internal stability is a basic requirement for a practical feedback system. This is because all interconnected systems may be unavoidably subject to some nonzero initial conditions and some (possibly small) errors, and it cannot be tolerated in practice that such errors at some locations will lead to unbounded signals at some other locations in the closed-loop system. Internal stability guarantees that all signals in a system are bounded provided that the injected signals (at any locations) are bounded. ♡

However, there are some special cases under which determining system stability is simple.

Corollary 5.4 *Suppose $\hat{K} \in \mathcal{RH}_\infty$. Then the system in Figure 5.2 is internally stable iff $(I - P\hat{K})^{-1}P \in \mathcal{RH}_\infty$.*

Proof. The necessity is obvious. To prove the sufficiency, it is sufficient to show that $(I - P\hat{K})^{-1} \in \mathcal{RH}_\infty$. But this follows from

$$(I - P\hat{K})^{-1} = I + (I - P\hat{K})^{-1}P\hat{K}$$

and $(I - P\hat{K})^{-1}P, \hat{K} \in \mathcal{RH}_\infty$. □

This corollary is in fact the basis for the classical control theory where the stability is checked only for one closed-loop transfer function with the implicit assumption that the controller itself is stable. Also, we have

Corollary 5.5 *Suppose $P \in \mathcal{RH}_\infty$. Then the system in Figure 5.2 is internally stable iff $\hat{K}(I - P\hat{K})^{-1} \in \mathcal{RH}_\infty$.*

Corollary 5.6 *Suppose $P \in \mathcal{RH}_\infty$ and $\hat{K} \in \mathcal{RH}_\infty$. Then the system in Figure 5.2 is internally stable iff $(I - P\hat{K})^{-1} \in \mathcal{RH}_\infty$.*

To study the more general case, define

$$\begin{aligned} n_c &:= \text{number of open rhp poles of } \hat{K}(s) \\ n_p &:= \text{number of open rhp poles of } P(s). \end{aligned}$$

Theorem 5.7 *The system is internally stable if and only if*

(i) the number of open rhp poles of $P(s)\hat{K}(s) = n_c + n_p$;

(ii) $\phi(s) := \det(I - P(s)\hat{K}(s))$ has all its zeros in the open left-half plane (i.e., $(I - P(s)\hat{K}(s))^{-1}$ is stable).

Proof. It is easy to show that $P\hat{K}$ and $(I - P\hat{K})^{-1}$ have the following realizations:

$$P\hat{K} = \left[\begin{array}{cc|c} A & B\hat{C} & B\hat{D} \\ 0 & \hat{A} & \hat{B} \\ \hline C & D\hat{C} & D\hat{D} \end{array}\right]$$

$$(I - P\hat{K})^{-1} = \left[\begin{array}{c|c} \bar{A} & \bar{B} \\ \hline \bar{C} & \bar{D} \end{array}\right]$$

where

$$\begin{aligned} \bar{A} &= \begin{bmatrix} A & B\hat{C} \\ 0 & \hat{A} \end{bmatrix} + \begin{bmatrix} B\hat{D} \\ \hat{B} \end{bmatrix} (I - D\hat{D})^{-1} \begin{bmatrix} C & D\hat{C} \end{bmatrix} \\ \bar{B} &= \begin{bmatrix} B\hat{D} \\ \hat{B} \end{bmatrix} (I - D\hat{D})^{-1} \\ \bar{C} &= (I - D\hat{D})^{-1} \begin{bmatrix} C & D\hat{C} \end{bmatrix} \\ \bar{D} &= (I - D\hat{D})^{-1}. \end{aligned}$$

It is also easy to see that $\bar{A} = \tilde{A}$. Hence, the system is internally stable iff $\bar{A}$ is stable.

Now suppose that the system is internally stable, then $(I - P\hat{K})^{-1} \in \mathcal{RH}_\infty$. This implies that all zeros of $\det(I - P(s)\hat{K}(s))$ must be in the left-half plane. So we only need to show that given condition (ii), condition (i) is necessary and sufficient for the internal stability. This follows by noting that $(\bar{A}, \bar{B})$ is stabilizable iff

$$\left(\begin{bmatrix} A & B\hat{C} \\ 0 & \hat{A} \end{bmatrix}, \begin{bmatrix} B\hat{D} \\ \hat{B} \end{bmatrix} \right) \tag{5.12}$$

is stabilizable; and $(\bar{C}, \bar{A})$ is detectable iff

$$\left(\begin{bmatrix} C & D\hat{C} \end{bmatrix}, \begin{bmatrix} A & B\hat{C} \\ 0 & \hat{A} \end{bmatrix} \right) \tag{5.13}$$

is detectable. But conditions (5.12) and (5.13) are equivalent to condition (i), i.e., $P\hat{K}$ has no unstable pole/zero cancelations. □

With this observation, the MIMO version of the Nyquist stability theorem is obvious.

Theorem 5.8 (Nyquist Stability Theorem) *The system is internally stable if and only if condition (i) in Theorem 5.7 is satisfied and the Nyquist plot of $\phi(j\omega)$ for $-\infty \leq \omega \leq \infty$ encircles the origin, $(0,0)$, $n_k + n_p$ times in the counter-clockwise direction.*

Proof. Note that by SISO Nyquist stability theorem, $\phi(s)$ has all zeros in the open left-half plane if and only if the Nyquist plot of $\phi(j\omega)$ for $-\infty \leq \omega \leq \infty$ encircles the origin, $(0,0)$, $n_k + n_p$ times in the counter-clockwise direction. □

5.4 Coprime Factorization over $\mathcal{RH}_\infty$

Recall that two polynomials $m(s)$ and $n(s)$, with, for example, real coefficients, are said to be *coprime* if their greatest common divisor is 1 (equivalent, they have no common zeros). It follows from Euclid's algorithm[1] that two polynomials m and n are coprime iff there exist polynomials $x(s)$ and $y(s)$ such that $xm + yn = 1$; such an equation is called a Bezout identity. Similarly, two transfer functions $m(s)$ and $n(s)$ in $\mathcal{RH}_\infty$ are said to be *coprime over* $\mathcal{RH}_\infty$ if there exists $x, y \in \mathcal{RH}_\infty$ such that

$$xm + yn = 1.$$

The more primitive, but equivalent, definition is that m and n are coprime if every common divisor of m and n is invertible in $\mathcal{RH}_\infty$, i.e.,

$$h, mh^{-1}, nh^{-1} \in \mathcal{RH}_\infty \Longrightarrow h^{-1} \in \mathcal{RH}_\infty.$$

More generally, we have

Definition 5.3 Two matrices M and N in $\mathcal{RH}_\infty$ are *right coprime over* $\mathcal{RH}_\infty$ if they have the same number of columns and if there exist matrices X_r and Y_r in $\mathcal{RH}_\infty$ such that

$$\begin{bmatrix} X_r & Y_r \end{bmatrix} \begin{bmatrix} M \\ N \end{bmatrix} = X_r M + Y_r N = I.$$

Similarly, two matrices $\tilde{M}$ and $\tilde{N}$ in $\mathcal{RH}_\infty$ are *left coprime over* $\mathcal{RH}_\infty$ if they have the same number of rows and if there exist matrices X_l and Y_l in $\mathcal{RH}_\infty$ such that

$$\begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix} \begin{bmatrix} X_l \\ Y_l \end{bmatrix} = \tilde{M} X_l + \tilde{N} Y_l = I.$$

Note that these definitions are equivalent to saying that the matrix $\begin{bmatrix} M \\ N \end{bmatrix}$ is left-invertible in $\mathcal{RH}_\infty$ and the matrix $\begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix}$ is right-invertible in $\mathcal{RH}_\infty$. These two equations are often called Bezout identities.

Now let P be a proper real-rational matrix. A *right-coprime factorization (rcf)* of P is a factorization $P = NM^{-1}$ where N and M are right-coprime over $\mathcal{RH}_\infty$. Similarly, a *left-coprime factorization (lcf)* has the form $P = \tilde{M}^{-1}\tilde{N}$ where $\tilde{N}$ and $\tilde{M}$ are left-coprime over $\mathcal{RH}_\infty$. A matrix $P(s) \in \mathcal{R}_p(s)$ is said to have *double coprime factorization* if there exist a right coprime factorization $P = NM^{-1}$, a left coprime factorization $P = \tilde{M}^{-1}\tilde{N}$, and X_r, Y_r, X_l, $Y_l \in \mathcal{RH}_\infty$ such that

$$\begin{bmatrix} X_r & Y_r \\ -\tilde{N} & \tilde{M} \end{bmatrix} \begin{bmatrix} M & -Y_l \\ N & X_l \end{bmatrix} = I. \tag{5.14}$$

Of course implicit in these definitions is the requirement that both M and $\tilde{M}$ be square and nonsingular.

[1] See, e.g., [Kailath, 1980, pp. 140-141].

Theorem 5.9 *Suppose $P(s)$ is a proper real-rational matrix and*

$$P = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$$

is a stabilizable and detectable realization. Let F and L be such that $A+BF$ and $A+LC$ are both stable, and define

$$\begin{bmatrix} M & -Y_l \\ N & X_l \end{bmatrix} = \left[\begin{array}{c|cc} A+BF & B & -L \\ \hline F & I & 0 \\ C+DF & D & I \end{array}\right] \tag{5.15}$$

$$\begin{bmatrix} X_r & Y_r \\ -\tilde{N} & \tilde{M} \end{bmatrix} = \left[\begin{array}{c|cc} A+LC & -(B+LD) & L \\ \hline F & I & 0 \\ C & -D & I \end{array}\right]. \tag{5.16}$$

Then $P = NM^{-1} = \tilde{M}^{-1}\tilde{N}$ are rcf *and* lcf*, respectively, and, furthermore, (5.14) is satisfied.*

Proof. The theorem follows by verifying the equation (5.14). □

Remark 5.2 Note that if P is stable, then we can take $X_r = X_l = I$, $Y_r = Y_l = 0$, $N = \tilde{N} = P$, $M = \tilde{M} = I$. ♡

Remark 5.3 The coprime factorization of a transfer matrix can be given a feedback control interpretation. For example, right coprime factorization comes out naturally from changing the control variable by a state feedback. Consider the state space equations for a plant P:

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du. \end{aligned}$$

Next, introduce a state feedback and change the variable

$$v := u - Fx$$

where F is such that $A+BF$ is stable. Then we get

$$\begin{aligned} \dot{x} &= (A+BF)x + Bv \\ u &= Fx + v \\ y &= (C+DF)x + Dv. \end{aligned}$$

Evidently from these equations, the transfer matrix from v to u is

$$M(s) = \left[\begin{array}{c|c} A+BF & B \\ \hline F & I \end{array}\right],$$

and that from v to y is

$$N(s) = \left[\begin{array}{c|c} A+BF & B \\ \hline C+DF & D \end{array}\right].$$

Therefore

$$u = Mv, \quad y = Nv$$

so that $y = NM^{-1}u$, i.e., $P = NM^{-1}$. ♡

We shall now see how coprime factorizations can be used to obtain alternative characterizations of internal stability conditions. Consider again the standard stability analysis diagram in Figure 5.2. We begin with any rcf's and lcf's of P and $\hat{K}$:

$$P = NM^{-1} = \tilde{M}^{-1}\tilde{N} \tag{5.17}$$

$$\hat{K} = UV^{-1} = \tilde{V}^{-1}\tilde{U}. \tag{5.18}$$

Lemma 5.10 *Consider the system in Figure 5.2. The following conditions are equivalent:*

1. *The feedback system is internally stable.*
2. $\begin{bmatrix} M & U \\ N & V \end{bmatrix}$ *is invertible in* $\mathcal{RH}_\infty$.
3. $\begin{bmatrix} \tilde{V} & -\tilde{U} \\ -\tilde{N} & \tilde{M} \end{bmatrix}$ *is invertible in* $\mathcal{RH}_\infty$.
4. $\tilde{M}V - \tilde{N}U$ *is invertible in* $\mathcal{RH}_\infty$.
5. $\tilde{V}M - \tilde{U}N$ *is invertible in* $\mathcal{RH}_\infty$.

Proof. As we saw in Lemma 5.3, internal stability is equivalent to

$$\begin{bmatrix} I & -\hat{K} \\ -P & I \end{bmatrix}^{-1} \in \mathcal{RH}_\infty$$

or, equivalently,

$$\begin{bmatrix} I & \hat{K} \\ P & I \end{bmatrix}^{-1} \in \mathcal{RH}_\infty. \tag{5.19}$$

Now

$$\begin{bmatrix} I & \hat{K} \\ P & I \end{bmatrix} = \begin{bmatrix} I & UV^{-1} \\ NM^{-1} & I \end{bmatrix} = \begin{bmatrix} M & U \\ N & V \end{bmatrix} \begin{bmatrix} M^{-1} & 0 \\ 0 & V^{-1} \end{bmatrix}$$

so that

$$\begin{bmatrix} I & \hat{K} \\ P & I \end{bmatrix}^{-1} = \begin{bmatrix} M & 0 \\ 0 & V \end{bmatrix} \begin{bmatrix} M & U \\ N & V \end{bmatrix}^{-1}.$$

Since the matrices

$$\begin{bmatrix} M & 0 \\ 0 & V \end{bmatrix}, \begin{bmatrix} M & U \\ N & V \end{bmatrix}$$

are right-coprime (this fact is left as an exercise for the reader), (5.19) holds iff

$$\begin{bmatrix} M & U \\ N & V \end{bmatrix}^{-1} \in \mathcal{RH}_\infty.$$

This proves the equivalence of conditions 1 and 2. The equivalence of 1 and 3 is proved similarly.

The conditions 4 and 5 are implied by 2 and 3 from the following equation:

$$\begin{bmatrix} \tilde{V} & -\tilde{U} \\ -\tilde{N} & \tilde{M} \end{bmatrix} \begin{bmatrix} M & U \\ N & V \end{bmatrix} = \begin{bmatrix} \tilde{V}M - \tilde{U}N & 0 \\ 0 & \tilde{M}V - \tilde{N}U \end{bmatrix}.$$

Since the left hand side of the above equation is invertible in $\mathcal{RH}_\infty$, so is the right hand side. Hence, conditions 4 and 5 are satisfied. We only need to show that either condition 4 or condition 5 implies condition 1. Let us show condition $5 \rightarrow 1$; this is obvious since

$$\begin{aligned} \begin{bmatrix} I & \hat{K} \\ P & I \end{bmatrix}^{-1} &= \begin{bmatrix} I & \tilde{V}^{-1}\tilde{U} \\ NM^{-1} & I \end{bmatrix}^{-1} \\ &= \begin{bmatrix} M & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} \tilde{V}M & \tilde{U} \\ N & I \end{bmatrix}^{-1} \begin{bmatrix} \tilde{V} & 0 \\ 0 & I \end{bmatrix} \in \mathcal{RH}_\infty \end{aligned}$$

if $\begin{bmatrix} \tilde{V}M & \tilde{U} \\ N & I \end{bmatrix}^{-1} \in \mathcal{RH}_\infty$ or if condition 5 is satisfied. □

Combining Lemma 5.10 and Theorem 5.9, we have the following corollary.

Corollary 5.11 *Let P be a proper real-rational matrix and $P = NM^{-1} = \tilde{M}^{-1}\tilde{N}$ be corresponding rcf and lcf over $\mathcal{RH}_\infty$. Then there exists a controller*

$$\hat{K}_0 = U_0 V_0^{-1} = \tilde{V}_0^{-1}\tilde{U}_0$$

with $U_0, V_0, \tilde{U}_0$, and $\tilde{V}_0$ in $\mathcal{RH}_\infty$ such that

$$\begin{bmatrix} \tilde{V}_0 & -\tilde{U}_0 \\ -\tilde{N} & \tilde{M} \end{bmatrix} \begin{bmatrix} M & U_0 \\ N & V_0 \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}. \tag{5.20}$$

Furthermore, let F and L be such that $A+BF$ and $A+LC$ are stable. Then a particular set of state space realizations for these matrices can be given by

$$\begin{bmatrix} M & U_0 \\ N & V_0 \end{bmatrix} = \left[\begin{array}{c|cc} A+BF & B & -L \\ \hline F & I & 0 \\ C+DF & D & I \end{array}\right] \tag{5.21}$$

$$\begin{bmatrix} \tilde{V}_0 & -\tilde{U}_0 \\ -\tilde{N} & \tilde{M} \end{bmatrix} = \left[\begin{array}{c|cc} A+LC & -(B+LD) & L \\ \hline F & I & 0 \\ C & -D & I \end{array}\right]. \tag{5.22}$$

Proof. The idea behind the choice of these matrices is as follows. Using the observer theory, find a controller $\hat{K}_0$ achieving internal stability; for example

$$\hat{K}_0 := \left[\begin{array}{c|c} A+BF+LC+LDF & -L \\ \hline F & 0 \end{array}\right]. \tag{5.23}$$

Perform factorizations

$$\hat{K}_0 = U_0 V_0^{-1} = \tilde{V}_0^{-1}\tilde{U}_0$$

which are analogous to the ones performed on P. Then Lemma 5.10 implies that each of the two left-hand side block matrices of (5.20) must be invertible in $\mathcal{RH}_\infty$. In fact, (5.20) is satisfied by comparing it with the equation (5.14). □

5.5 Feedback Properties

In this section, we discuss the properties of a feedback system. In particular, we consider the benefit of the feedback structure and the concept of design tradeoffs for conflicting objectives – namely, how to achieve the benefits of feedback in the face of uncertainties.

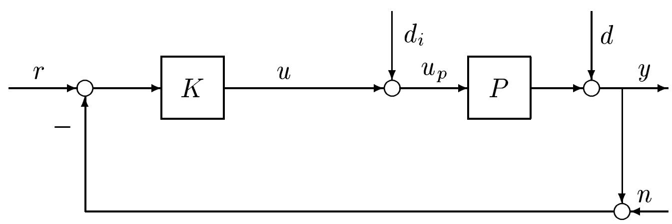

Figure 5.3: Standard Feedback Configuration

Consider again the feedback system shown in Figure 5.1. For convenience, the system diagram is shown again in Figure 5.3. For further discussion, it is convenient to define the *input loop transfer matrix*, L_i, and *output loop transfer matrix*, L_o, as

$$L_i = KP, \quad L_o = PK,$$

respectively, where L_i is obtained from breaking the loop at the input (u) of the plant while L_o is obtained from breaking the loop at the output (y) of the plant. The *input sensitivity* matrix is defined as the transfer matrix from d_i to u_p:

$$S_i = (I+L_i)^{-1}, \quad u_p = S_i d_i.$$

And the *output sensitivity* matrix is defined as the transfer matrix from d to y:

$$S_o = (I+L_o)^{-1}, \quad y = S_o d.$$

The *input* and *output complementary sensitivity* matrices are defined as

$$T_i = I - S_i = L_i(I + L_i)^{-1}$$

$$T_o = I - S_o = L_o(I + L_o)^{-1},$$

respectively. (The word *complementary* is used to signify the fact that T is the complement of S, $T = I - S$.) The matrix $I + L_i$ is called *input return difference matrix* and $I + L_o$ is called *output return difference matrix.*

It is easy to see that the closed-loop system, if it is internally stable, satisfies the following equations:

$$y = T_o(r - n) + S_oPd_i + S_od \tag{5.24}$$

$$r - y = S_o(r - d) + T_on - S_oPd_i \tag{5.25}$$

$$u = KS_o(r - n) - KS_od - T_id_i \tag{5.26}$$

$$u_p = KS_o(r - n) - KS_od + S_id_i. \tag{5.27}$$

These four equations show the fundamental benefits and design objectives inherent in feedback loops. For example, equation (5.24) shows that the effects of disturbance d on the plant output can be made "small" by making the output sensitivity function S_o small. Similarly, equation (5.27) shows that the effects of disturbance d_i on the plant input can be made small by making the input sensitivity function S_i small. The notion of smallness for a transfer matrix in a certain range of frequencies can be made explicit using frequency dependent singular values, for example, $\overline{\sigma}(S_o) < 1$ over a frequency range would mean that the effects of disturbance d at the plant output are effectively desensitized over that frequency range.

Hence, good disturbance rejection at the plant output (y) would require that

$$\begin{aligned} \overline{\sigma}(S_o) &= \overline{\sigma}\left((I + PK)^{-1}\right) = \frac{1}{\underline{\sigma}(I + PK)}, \quad \text{(for disturbance at plant output, } d) \\ \overline{\sigma}(S_oP) &= \overline{\sigma}\left((I + PK)^{-1}P\right) = \overline{\sigma}(PS_i), \quad \text{(for disturbance at plant input, } d_i) \end{aligned}$$

be made small and good disturbance rejection at the plant input (u_p) would require that

$$\begin{aligned} \overline{\sigma}(S_i) &= \overline{\sigma}\left((I + KP)^{-1}\right) = \frac{1}{\underline{\sigma}(I + KP)}, \quad \text{(for disturbance at plant input, } d_i) \\ \overline{\sigma}(S_iK) &= \overline{\sigma}\left(K(I + PK)^{-1}\right) = \overline{\sigma}(KS_o), \quad \text{(for disturbance at plant output, } d) \end{aligned}$$

be made small, particularly in the low frequency range where d and d_i are usually significant.

Note that

$$\begin{aligned} \underline{\sigma}(PK) - 1 &\leq \underline{\sigma}(I + PK) \leq \underline{\sigma}(PK) + 1 \\ \underline{\sigma}(KP) - 1 &\leq \underline{\sigma}(I + KP) \leq \underline{\sigma}(KP) + 1 \end{aligned}$$

then

$$\frac{1}{\underline{\sigma}(PK)+1} \leq \overline{\sigma}(S_o) \leq \frac{1}{\underline{\sigma}(PK)-1}, \quad \text{if} \quad \underline{\sigma}(PK) > 1$$
$$\frac{1}{\underline{\sigma}(KP)+1} \leq \overline{\sigma}(S_i) \leq \frac{1}{\underline{\sigma}(KP)-1}, \quad \text{if} \quad \underline{\sigma}(KP) > 1.$$

These equations imply that

$$\overline{\sigma}(S_o) \ll 1 \iff \underline{\sigma}(PK) \gg 1$$
$$\overline{\sigma}(S_i) \ll 1 \iff \underline{\sigma}(KP) \gg 1.$$

Now suppose P and K are invertible, then

$$\underline{\sigma}(PK) \gg 1 \text{ or } \underline{\sigma}(KP) \gg 1 \iff \overline{\sigma}(S_oP) = \overline{\sigma}\left((I+PK)^{-1}P\right) \approx \overline{\sigma}(K^{-1}) = \frac{1}{\underline{\sigma}(K)}$$
$$\underline{\sigma}(PK) \gg 1 \text{ or } \underline{\sigma}(KP) \gg 1 \iff \overline{\sigma}(KS_o) = \overline{\sigma}\left(K(I+PK)^{-1}\right) \approx \overline{\sigma}(P^{-1}) = \frac{1}{\underline{\sigma}(P)}.$$

Hence good performance at plant output (y) requires in general large output loop gain $\underline{\sigma}(L_o) = \underline{\sigma}(PK) \gg 1$ in the frequency range where d is significant for desensitizing d and large enough controller gain $\underline{\sigma}(K) \gg 1$ in the frequency range where d_i is significant for desensitizing d_i. Similarly, good performance at plant input (u_p) requires in general large input loop gain $\underline{\sigma}(L_i) = \underline{\sigma}(KP) \gg 1$ in the frequency range where d_i is significant for desensitizing d_i and large enough plant gain $\underline{\sigma}(P) \gg 1$ in the frequency range where d is significant, which can not changed by controller design, for desensitizing d. (It should be noted that in general $S_o \neq S_i$ unless K and P are square and diagonal which is true if P is a scalar system. Hence, small $\overline{\sigma}(S_o)$ does not necessarily imply small $\overline{\sigma}(S_i)$; in other words, good disturbance rejection at the output does not necessarily mean good disturbance rejection at the plant input.)

Hence, *good multivariable feedback loop design boils down to achieving high loop (and possibly controller) gains in the necessary frequency range.*

Despite the simplicity of this statement, feedback design is by no means trivial. This is true because loop gains cannot be made arbitrarily high over arbitrarily large frequency ranges. Rather, they must satisfy certain performance tradeoff and design limitations. A major performance tradeoff, for example, concerns commands and disturbance error reduction versus stability under the model uncertainty. Assume that the plant model is perturbed to $(I+\Delta)P$ with Δ stable, and assume that the system is nominally stable, i.e., the closed-loop system with $\Delta = 0$ is stable. Now the perturbed closed-loop system is stable if

$$\det\left(I + (I+\Delta)PK\right) = \det(I+PK)\det(I+\Delta T_o)$$

has no right-half plane zero. This would in general amount to requiring that $\|\Delta T_o\|$ be small or that $\bar{\sigma}(T_o)$ be small at those frequencies where Δ is significant, typically at high frequency range, which in turn implies that the loop gain, $\overline{\sigma}(L_o)$, should be small at those frequencies.

Still another tradeoff is with the sensor noise error reduction. The conflict between the disturbance rejection and the sensor noise reduction is evident in equation (5.24). Large $\underline{\sigma}(L_o(j\omega))$ values over a large frequency range make errors due to d small. However, they also make errors due to n large because this noise is "passed through" over the same frequency range, i.e.,

$$y = T_o(r-n) + S_oPd_i + S_od \approx (r-n).$$

Note that n is typically significant in the high frequency range. Worst still, large loop gains outside of the bandwidth of P, i.e., $\underline{\sigma}(L_o(j\omega)) \gg 1$ or $\underline{\sigma}(L_i(j\omega)) \gg 1$ while $\overline{\sigma}(P(j\omega)) \ll 1$, can make the control activity (u) quite unacceptable, which may cause the saturation of actuators. This follows from

$$u = KS_o(r-n-d) - T_id_i = S_iK(r-n-d) - T_id_i \approx P^{-1}(r-n-d) - d_i.$$

Here, we have assumed P to be square and invertible for convenience. The resulting equation shows that disturbances and sensor noise are actually amplified at u whenever the frequency range significantly exceeds the bandwidth of P, since for ω such that $\overline{\sigma}(P(j\omega)) \ll 1$, we have

$$\underline{\sigma}[P^{-1}(j\omega)] = \frac{1}{\overline{\sigma}[P(j\omega)]} \gg 1.$$

Similarly, the controller gain, $\overline{\sigma}(K)$, should also be kept not too large in the frequency range where the loop gain is small in order to not saturate the actuators. This is because for small loop gain $\overline{\sigma}(L_o(j\omega)) \ll 1$ or $\overline{\sigma}(L_i(j\omega)) \ll 1$

$$u = KS_o(r-n-d) - T_id_i \approx K(r-n-d).$$

Therefore, it is desirable to keep $\overline{\sigma}(K)$ not too large when the loop gain is small.

To summarize the above discussion, we note that good performance requires in some frequency range, typically some low frequency range $(0, \omega_l)$:

$$\underline{\sigma}(PK) \gg 1, \quad \underline{\sigma}(KP) \gg 1, \quad \underline{\sigma}(K) \gg 1$$

and good robustness and good sensor noise rejection require in some frequency range, typically some high frequency range (ω_h, ∞)

$$\overline{\sigma}(PK) \ll 1, \quad \overline{\sigma}(KP) \ll 1, \quad \overline{\sigma}(K) \leq M$$

where M is not too large. These design requirements are shown graphically in Figure 5.4. The specific frequencies ω_l and ω_h depend on the specific applications and the knowledge one has on the disturbance characteristics, the modeling uncertainties, and the sensor noise levels.

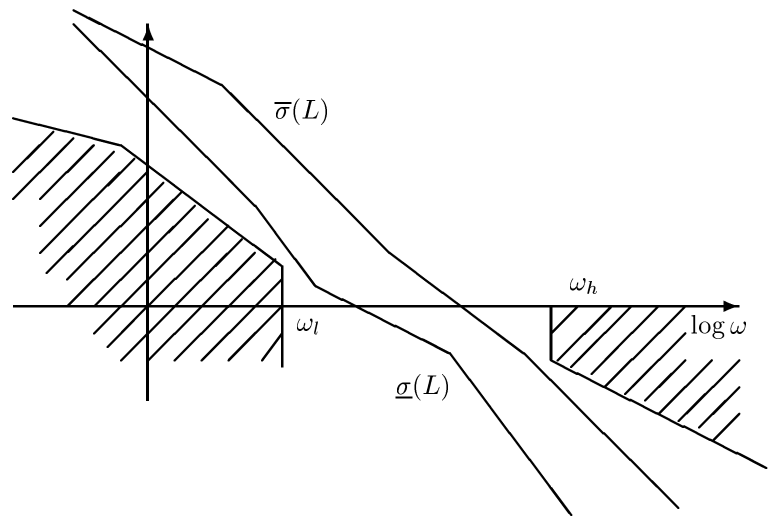

Figure 5.4: Desired Loop Gain

5.6 The Concept of Loop Shaping

The analysis in the last section motivates a conceptually simple controller design technique: loop shaping. Loop shaping controller design involves essentially finding a controller K that shapes the loop transfer function L so that the loop gains, $\underline{\sigma}(L)$ and $\overline{\sigma}(L)$, clear the boundaries specified by the performance requirements at low frequencies and by the robustness requirements at high frequencies as shown in Figure 5.4.

In the SISO case, the loop shaping design technique is particularly effective and simple since $\overline{\sigma}(L) = \underline{\sigma}(L) = |L|$. The design procedure can be completed in two steps:

SISO Loop Shaping

(1) Find a rational strictly proper transfer function L which contains all the right half plane poles and zeros of P such that $|L|$ clears the boundaries specified by the performance requirements at low frequencies and by the robustness requirements at high frequencies as shown in Figure 5.4.

L must also be chosen so that $1+L$ has all zeros in the open left half plane, which can usually be guaranteed by making L well-behaved in the crossover region, i.e., L should not be decreasing too fast in the frequency range of $|L(j\omega)| \approx 1$.

(2) The controller is given by $K = L/P$.

The loop shaping for MIMO system can be done similarly if the singular values of the loop transfer functions are used for the loop gains.

MIMO Loop Shaping

(1) Find a rational strictly proper transfer function L which contains all the right half plane poles and zeros of P so that the product of P and $P^{-1}L$ (or LP^{-1}) has no unstable poles and/or zeros cancelations, and $\underline{\sigma}(L)$ clears the boundary specified by the performance requirements at low frequencies and $\overline{\sigma}(L)$ clears the boundary specified by the robustness requirements at high frequencies as shown in Figure 5.4.

L must also be chosen so that $\det(I+L)$ has all zeros in the open left half plane. (This is not easy for MIMO systems.)

(2) The controller is given by $K = P^{-1}L$ if L is the output loop transfer function (or $K = LP^{-1}$ if L is the input loop transfer function).

The loop shaping design technique can be quite useful especially for SISO control system design. However, there are severe limitations when it is used for MIMO system design.

Limitations of MIMO Loop Shaping

Although the above loop shaping design can be effective in some of applications, there are severe intrinsic limitations. Some of these limitations are listed below:

- The loop shaping technique described above can only effectively deal with problems with uniformly specified performance and robustness specifications. More specifically, the method can not effectively deal with problems with different specifications in different channels and/or problems with different uncertainty characteristics in different channels without introducing significant conservatism. To illustrate this difficulty, consider an uncertain dynamical system

$$P_\Delta = (I+\Delta)P$$

where P is the nominal plant and Δ is the multiplicative modeling error. Assume that Δ can be written in the following form

$$\Delta = \tilde{\Delta}W_t, \quad \overline{\sigma}(\tilde{\Delta}) < 1.$$

Then, for robust stability, we would require $\overline{\sigma}(\Delta T_o) = \overline{\sigma}(\tilde{\Delta}W_t T_o) < 1$ or $\overline{\sigma}(W_t T_o) \le 1$. If a uniform bound is required on the loop gain to apply the loop shaping technique, we would need to overbound $\overline{\sigma}(W_t T_o)$:

$$\overline{\sigma}(W_t T_o) \le \overline{\sigma}(W_t)\overline{\sigma}(T_o) \le \overline{\sigma}(W_t)\frac{\overline{\sigma}(L_o)}{1-\overline{\sigma}(L_o)}, \quad \text{if} \quad \overline{\sigma}(L_o) < 1$$

and the robust stability requirement is implied by

$$\overline{\sigma}(L_o) \le \frac{1}{\overline{\sigma}(W_t)+1} \approx \frac{1}{\overline{\sigma}(W_t)}, \quad \text{if} \quad \overline{\sigma}(L_o) < 1.$$

Similarly, if the performance requirements, say output disturbance rejection, are not uniformly specified in all channels but by a weighting matrix W_s such that $\overline{\sigma}(W_s S_o) \leq 1$, then it is also necessary to overbound $\overline{\sigma}(W_s S_o)$ in order to apply the loop shaping techniques:

$$\overline{\sigma}(W_s S_o) \leq \overline{\sigma}(W_s)\overline{\sigma}(S_o) \leq \frac{\overline{\sigma}(W_s)}{\underline{\sigma}(L_o) - 1}, \quad \text{if} \quad \underline{\sigma}(L_o) > 1$$

and the performance requirement is implied by

$$\underline{\sigma}(L_o) \geq \overline{\sigma}(W_s) + 1 \approx \overline{\sigma}(W_s), \quad \text{if} \quad \underline{\sigma}(L_o) > 1.$$

It is possible that the bounds for the loop shape may contradict each other at some frequency range, as shown in the figure 5.5. However, this does not imply that there is no controller that will satisfy both nominal performance and robust stability except for SISO systems. This contradiction happens because the bounds

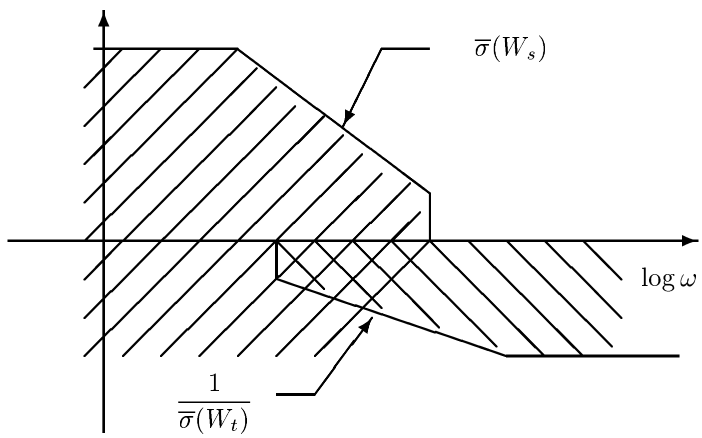

Figure 5.5: Conflict Requirements

do not utilize the structure of weights, W_s and W_t; and the bounds are only sufficient conditions for robust stability and nominal performance. This possibility can be further illustrated by the following example: Assume that a two-input and two-output system transfer matrix is given by

$$P(s) = \frac{1}{s+1} \begin{bmatrix} 1 & \alpha \\ 0 & 1 \end{bmatrix},$$

and suppose the weighting matrices are given by

$$W_s = \begin{bmatrix} \frac{1}{s+1} & \frac{\alpha}{(s+1)(s+2)} \\ 0 & \frac{1}{s+1} \end{bmatrix}, \quad W_t = \begin{bmatrix} \frac{s+2}{s+10} & \frac{\alpha(s+1)}{s+10} \\ 0 & \frac{s+2}{s+10} \end{bmatrix}.$$

It is easy to show that for large α, the weighting functions are as shown in Figure 5.5, and thus the above loop shaping technique cannot be applied. However, it is also easy to show that the system with controller

$$K = I_2$$

gives

$$W_s S = \begin{bmatrix} \frac{1}{s+2} & 0 \\ 0 & \frac{1}{s+2} \end{bmatrix}, \quad W_t T = \begin{bmatrix} \frac{1}{s+10} & 0 \\ 0 & \frac{1}{s+10} \end{bmatrix}$$

and, therefore, the robust performance criterion is satisfied.

- Even if all of the above problems can be avoided, it may still be difficult to find a *matrix function* L_o so that $K = P^{-1}L_o$ is stabilizing. This becomes much harder if P is non-minimum phase and/or unstable.

Hence some new methodologies have to be introduced to solve complicated problems. The so-called LQG/LTR (Linear Quadratic Gaussian/Loop Transfer Recovery) procedure developed first by Doyle and Stein [1981] and extended later by various authors can solve some of the problems, but it is essentially limited to nominally minimum phase and output multiplicative uncertain systems. For these reasons, it will not be introduced here. This motivates us to consider the closed-loop performance directly in terms of the closed-loop transfer functions instead of open loop transfer functions. The following section considers some simple closed-loop performance problem formulations.

5.7 Weighted $\mathcal{H}_2$ and $\mathcal{H}_\infty$ Performance

In this section, we consider how to formulate some performance objectives into mathematically tractable problems. As shown in section 5.5, the performance objectives of a feedback system can usually be specified in terms of requirements on the sensitivity functions and/or complementary sensitivity functions or in terms of some other closed-loop transfer functions. For instance, the performance criteria for a scalar system may be specified as requiring

$$\begin{cases} |s(j\omega)| \leq \alpha < 1 & \forall \omega \leq \omega_0, \\ |s(j\omega)| \leq \beta > 1 & \forall \omega > \omega_0 \end{cases}$$

where $s(j\omega) = \frac{1}{1+p(j\omega)k(j\omega)}$. However, it is much more convenient to reflect the system performance objectives by choosing appropriate weighting functions. For example, the above performance objective can be written as

$$|w_s(j\omega)s(j\omega)| \leq 1, \quad \forall \omega$$

with

$$|w_s(j\omega)| = \begin{cases} \alpha^{-1} & \forall \omega \leq \omega_0, \\ \beta^{-1} & \forall \omega > \omega_0. \end{cases}$$

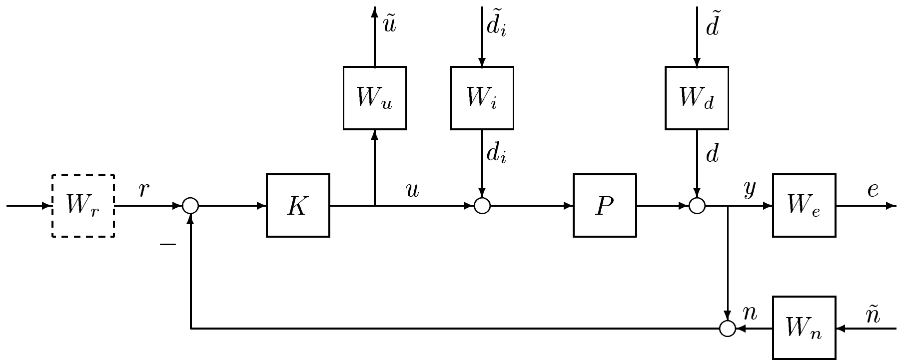

Figure 5.6: Standard Feedback Configuration with Weights

In order to use w_s in control design, a rational transfer function w_s is usually used to approximate the above frequency response.

The advantage of using weighted performance specifications is obvious in multivariable system design. First of all, some components of a vector signal are usually more important than others. Secondly, each component of the signal may not be measured in the same metric; for example, some components of the output error signal may be measured in terms of length, and the others may be measured in terms of voltage. Therefore, weighting functions are essential to make these components comparable. Also, we might be primarily interested in rejecting errors in a certain frequency range (for example low frequencies), hence some frequency dependent weights must be chosen.

In general, we shall modify the standard feedback diagram in Figure 5.3 into Figure 5.6. The weighting functions in Figure 5.6 are chosen to reflect the design objectives and knowledge on the disturbances and sensor noise. For example, W_d and W_i may be chosen to reflect the frequency contents of the disturbances d and d_i or they may be used to model the disturbance power spectrum depending on the nature of signals involved in the practical systems. The weighting matrix W_n is used to model the frequency contents of the sensor noise while W_e may be used to reflect the requirements on the shape of certain closed-loop transfer functions, for example, the shape of the output sensitivity function. Similarly, W_u may be used to reflect some restrictions on the control or actuator signals, and the dashed precompensator W_r is an optional element used to achieve deliberate command shaping or to represent a non-unity feedback system in equivalent unity feedback form.

It is, in fact, essential that some appropriate weighting matrices be used in order to utilize the optimal control theory discussed in this book, i.e., $\mathcal{H}_2$ and $\mathcal{H}_\infty$ theory. So a very important step in controller design process is to choose the appropriate weights, W_e, W_d, W_u, and possibly W_n, W_i, and W_r. The appropriate choice of weights for a particular practical problem is not trivial. In many occasions, as in the scalar case, the weights are chosen purely as a design parameter without any physical bases, so these weights may be treated as tuning parameters which are chosen by the designer to achieve the best compromise between the conflicting objectives. The selection of the weighting matrices should be guided by the expected system inputs and the relative importance of the outputs.

Hence, control design may be regarded as a process of choosing a controller K such that certain weighted signals are made small in some sense. There are many different ways to define the smallness of a signal or transfer matrix, as we have discussed in the last chapter. Different definitions lead to different control synthesis methods, and some are much harder than others. A control engineer should make a judgment of the mathematical complexity versus engineering requirements.

Below, we introduce two classes of performance formulations: $\mathcal{H}_2$ and $\mathcal{H}_\infty$ criteria. For the simplicity of presentation, we shall assume $d_i = 0$ and $n = 0$.

$\mathcal{H}_2$ Performance

Assume, for example, that the disturbance $\tilde{d}$ can be approximately modeled as an impulse with random input direction, i.e.,

$$\tilde{d}(t) = \eta\delta(t)$$

and

$$E(\eta\eta^*) = I$$

where E denotes the expectation. We may choose to minimize the expected energy of the error e due to the disturbance $\tilde{d}$:

$$E\left\{\|e\|_2^2\right\} = E\left\{\int_0^\infty \|e\|^2\,dt\right\} = \|W_e S_o W_d\|_2^2.$$

Alternatively, if we suppose that the disturbance $\tilde{d}$ can be approximately modeled as white noise with $S_{\tilde{d}\tilde{d}} = I$, then

$$S_{ee} = (W_e S_o W_d) S_{\tilde{d}\tilde{d}} (W_e S_o W_d)^*,$$

and we may chose to minimize the power of e:

$$\|e\|_P^2 = \frac{1}{2\pi}\int_{-\infty}^{\infty} \operatorname{Trace} S_{ee}(j\omega)\;d\omega = \|W_e S_o W_d\|_2^2.$$

In general, a controller minimizing only the above criterion can lead to a very large control signal u that could cause saturation of the actuators as well as many other undesirable problems. Hence, for a realistic controller design, it is necessary to include the control signal u in the penalty function. Thus, our design criterion would usually be something like this

$$E\left\{\|e\|_2^2 + \rho^2\|\tilde{u}\|_2^2\right\} = \left\| \begin{bmatrix} W_e S_o W_d \\ \rho W_u K S_o W_d \end{bmatrix} \right\|_2^2$$

with some appropriate choice of weighting matrix W_u and scalar ρ. The parameter ρ clearly defines the tradeoff we discussed earlier between good disturbance rejection at the output and control effort (or disturbance and sensor noise rejection at the actuators). Note that ρ can be set to $\rho = 1$ by an appropriate choice of W_u. This problem can be viewed as minimizing the *energy* consumed by the system in order to reject the disturbance d.

This type of problem was the dominant paradigm in the 1960's and 1970's and is usually referred to as Linear Quadratic Gaussian Control or simply as LQG. (They will also be referred to as $\mathcal{H}_2$ mixed sensitivity problems for the consistency with the $\mathcal{H}_\infty$ problems discussed next.) The development of this paradigm stimulated extensive research efforts and is responsible for important technological innovation, particularly in the area of estimation. The theoretical contributions include a deeper understanding of linear systems and improved computational methods for complex systems through state-space techniques. The major limitation of this theory is the lack of formal treatment of uncertainty in the plant itself. By allowing only additive noise for uncertainty, the stochastic theory ignored this important practical issue. Plant uncertainty is particularly critical in feedback systems.

$\mathcal{H}_\infty$ Performance

Although the $\mathcal{H}_2$ norm (or $\mathcal{L}_2$ norm) may be a meaningful performance measure and although LQG theory can give efficient design compromises under certain disturbance and plant assumptions, the $\mathcal{H}_2$ norm suffers a major deficiency. This deficiency is due to the fact that the tradeoff between disturbance error reduction and sensor noise error reduction is not the only constraint on feedback design. The problem is that these performance tradeoffs are often overshadowed by a second limitation on high loop gains – namely, the requirement for tolerance to uncertainties. Though a controller may be designed using FDLTI models, the design must be implemented and operated with a real physical plant. The properties of physical systems, in particular the ways in which they deviate from finite-dimensional linear models, put strict limitations on the frequency range over which the loop gains may be large.

A solution to this problem would be to put explicit constraints on the loop gain in the penalty function. For instance, one may chose to minimize

$$\sup_{\|\tilde{d}\|_2 \leq 1} \|e\|_2 = \|W_e S_o W_d\|_\infty ,$$

subject to some restrictions on the control energy or control bandwidth:

$$\sup_{\|\tilde{d}\|_2 \leq 1} \|\tilde{u}\|_2 = \|W_u K S_o W_d\|_\infty .$$

Or more frequently, one may introduce a parameter ρ and a mixed criterion

$$\sup_{\|\tilde{d}\|_2 \leq 1} \left\{ \|e\|_2^2 + \rho^2 \|\tilde{u}\|_2^2 \right\} = \left\| \begin{bmatrix} W_e S_o W_d \\ \rho W_u K S_o W_d \end{bmatrix} \right\|_\infty^2 .$$

This problem can also be regarded as minimizing the maximum power of the error subject to all bounded power disturbances: let

$$\hat{e} := \begin{bmatrix} e \\ \tilde{u} \end{bmatrix}$$

then

$$S_{\hat{e}\hat{e}} = \begin{bmatrix} W_e S_o W_d \\ W_u K S_o W_d \end{bmatrix} S_{\tilde{d}\tilde{d}} \begin{bmatrix} W_e S_o W_d \\ W_u K S_o W_d \end{bmatrix}^*$$

and

$$\sup_{\|\tilde{d}\|_P \le 1} \|\hat{e}\|_P^2 = \sup_{\|\tilde{d}\|_P \le 1} \left\{ \frac{1}{2\pi} \int_{-\infty}^{\infty} \operatorname{Trace} S_{\hat{e}\hat{e}}(j\omega)\ d\omega \right\} = \left\| \begin{bmatrix} W_e S_o W_d \\ W_u K S_o W_d \end{bmatrix} \right\|_\infty^2 .$$

Alternatively, if the system robust stability margin is the major concern, the weighted complementary sensitivity has to be limited. Thus the whole cost function may be

$$\left\| \begin{bmatrix} W_e S_o W_d \\ \rho W_1 T_o W_2 \end{bmatrix} \right\|_\infty$$

where W_1 and W_2 are the frequency dependent uncertainty scaling matrices. These design problems are usually called $\mathcal{H}_\infty$ mixed sensitivity problems. For a scalar system, an $\mathcal{H}_\infty$ norm minimization problem can also be viewed as minimizing the maximum magnitude of the system's steady-state response with respect to the worst case sinusoidal inputs.

5.8 Notes and References

The presentation of this chapter is based primarily on Doyle [1984]. The discussion of internal stability and coprime factorization can also be found in Francis [1987] and Vidyasagar [1985]. The loop shaping design is well known for SISO systems in the classical control theory. The idea was extended to MIMO systems by Doyle and Stein [1981] using LQG design technique. The limitations of the loop shaping design are discussed in detail in Stein and Doyle [1991]. Chapter 18 presents another loop shaping method using $\mathcal{H}_\infty$ control theory which has the potential to overcome the limitations of the LQG/LTR method.